

Assignment 5

(1)

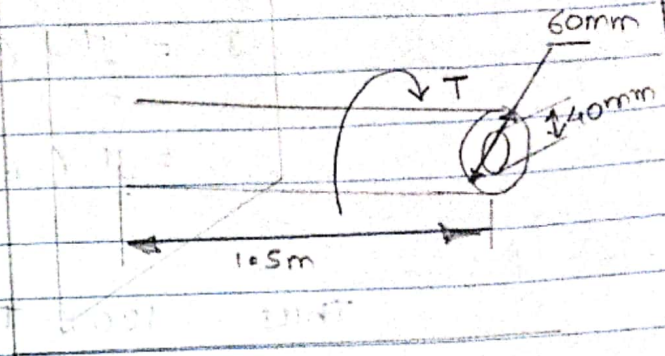
Given:

hollow cylindrical shaft

$$(L) = 1.5 \text{ m}$$

$$(d)_i = 40 \text{ mm}$$

$$(d)_o = 60 \text{ mm}$$



(a) Critical section in a shaft under torsion is located at section with smallest Diameter. But as this is hollow shaft, critical section would occur at point of max stress, i.e. outer surface

Outer Diameter = 60 mm

(b) Critical point is farthest from centre of shaft i.e. outer surface = 60 mm.

(c) Shear stress in circular shaft under torsion

$$\tau = \frac{T \cdot r_o}{J}$$

$$120 = \frac{T \cdot 0.03}{J}$$

(J → Polar moment of Inertia)

$$\tau_{min} = \frac{1633.628 \times 10^{-7}}{20.420 \times 10^{-7}}$$

$$\tau_{min} = 80 \text{ MPa}$$

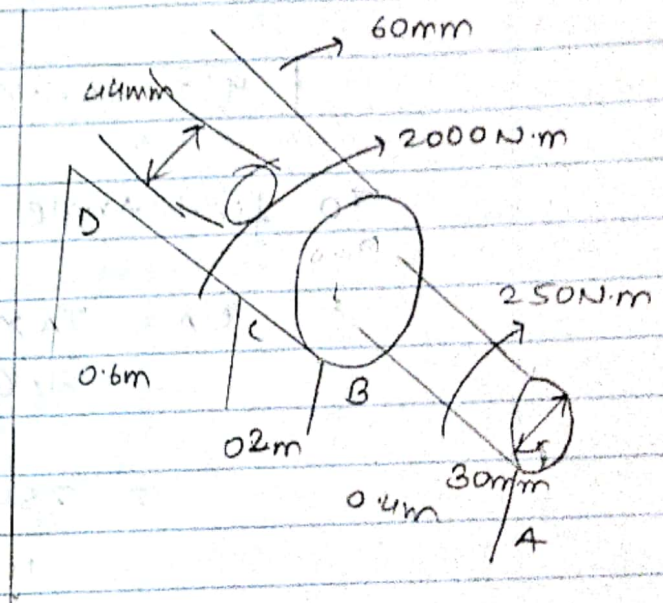
(a) Largest Torque (T) = ~~408.407×10^5~~
 4084.07 Nm

(b) Min Torque (T_{min}) = 80 MPa

Q2

(a) The critical section is where 44mm hole is drilled.

Critical point is at surface of drilled section particularly at edge of hole.



(b) Polar moment of Inertia

$$J = \frac{\pi}{2} ((r_o)^4 - (r_i)^4)$$

$$J = \frac{\pi}{2} [(0.03)^4 - (0.022)^4]$$

$$J = 1.0145 \times 10^{-7}$$

Thus Torsional shear stress
is given by

$$\tau = \frac{T \cdot r}{J}$$

$$T = 2000 \text{ Nm} \quad (\text{Given})$$

$$r = 0.03 \text{ m}$$

$$J = 1.0145 \times 10^{-7} \text{ m}^4$$

$$\tau = 66.3 \text{ MPa}$$

To find Angle of twist

$$\theta_A = \frac{T \times L}{J \cdot G}$$

$$= \frac{250 \times 0.4}{1.0145 \times 10^{-7} \times 77 \times 10^9}$$

$$\theta_A = 75.8994$$

$$\theta_{\text{degree}} = 75.8994$$

Q8]

(a) Max and Min shear at BC.

For hollow shaft

Shear due to torque

$$\tau = \frac{T r}{J}$$

$$J = \frac{\pi}{2} (r_o^4 - r_i^4)$$

$$\tau = \frac{26 \times 10^3 \times 0.06}{J}$$

$$J = \frac{\pi}{2} [(0.06)^4 - (0.045)^4]$$

$$= \frac{3120}{[8.85 \times 10^{-6}]}$$

$$= \frac{3120}{1.39 \times 10^{-5}}$$

$$\tau_{max} = 112.10 \text{ MPa}$$

For τ_{min} , we have

$$\tau_{min} = \frac{T r_i}{J} = \frac{26 \times 10^3 \times 0.045}{1.39 \times 10^{-5}}$$

$$\tau_{min} = 84.07 \text{ MPa}$$

Req diameter of shaft AB and CD

$$\text{Shear stress} = \tau = \frac{T \cdot r}{J}$$

$$J = \frac{\pi}{2} r^4$$

$$r = \left(\frac{T}{\tau_{allowed} \cdot \frac{\pi}{2}} \right)^{1/3}$$

$$= \left(\frac{26 \times 10^3}{65 \times 10 \times \frac{\pi}{2}} \right)^{1/3}$$

$$= (57.99)^{1/3}$$

$$= (60)^{1/3}$$

$$= 3.91$$

diameter =

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Req

(Just for Explanation)

4) Note: Shaft C is hollow shaft. Thus torque acting on shaft B (14 kNm) is the governing torque of solid shaft AB and CD.

By Torsional shear stress formula

$$\tau = \frac{T \cdot r}{J}$$

J for solid circular shaft

$$J = \frac{\pi d^4}{32}$$

Thus By substituting

$$\tau = \frac{T \cdot d/2}{\frac{\pi d^4}{32}}$$

$$\tau = \frac{16 \cdot T}{\pi d^3}$$

Max allowed stress

$$65 \times 10^6 = \frac{16 \cdot 14000}{\pi \times d^3}$$

$$d^3 = 0.001097$$

$$d = 0.1034 \text{ m}$$

$$d = 103.4 \text{ mm}$$

Req Diameter of shaft is 103.4 mm

Q4.

Given:

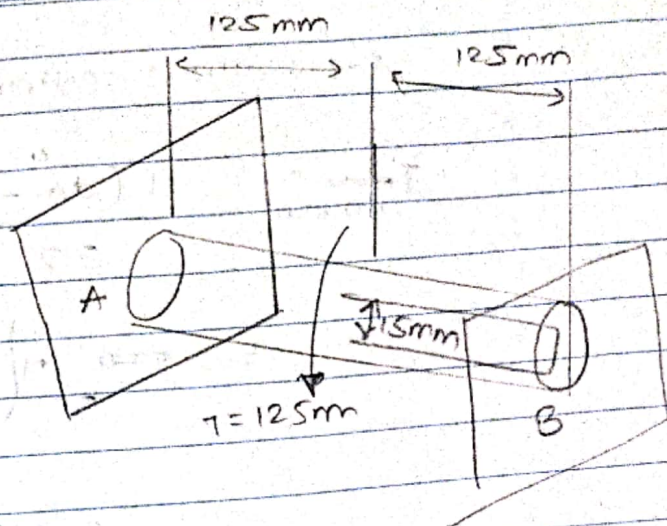
Shaft AB of length = 250mm

Diameter = 25mm

Cavity = 125mm

Diameter = 15mm

Torque = 125 Nm at midsection



(A)

Torque at support A and B (T_A and T_B)

$$T_A = T_B = \frac{T}{2} = \frac{125}{2} = 62.5 \text{ Nm}$$

Critical Pt for hollow section - 12.5mm Radius

(B) Critical sect will be at Junction where Diameter of shaft changes. (125mm from B)

Critical Point - Outer surface of shaft.

(C) Polar Moment of Inertia (J) for solid & circular section

$$J = \frac{\pi d^4}{32}$$

$$J = \frac{\pi (25 \times 10^{-3})^4}{32} = 3.067 \times 10^{-9} \text{ mm}^4$$

for hollow section

$$I_{\text{max}} = \frac{\pi (d_o^4 - d_i^4)}{32}$$
$$= \frac{\pi [(25 \times 10^{-3})^4 - (15 \times 10^{-3})^4]}{32}$$
$$= 2.418 \times 10^{-9} \text{ m}^4$$

To find shear stress at critical point section

$$\tau_{\text{max}} = \frac{T \cdot r}{J}$$

$$\tau_{\text{solid}} = \frac{62.5 \times 0.0125}{2.067 \times 10^{-9}} = 255.16 \text{ kPa}$$

$$\tau_{\text{hollow}} = \frac{62.5 \times 0.0125}{2.418 \times 10^{-9}} = 323.76 \text{ kPa}$$

Thus state of stress at critical section
has max shear stress of 323.76 kPa

(1) ~~Locate~~ Critical sec is where Torque^s applied and two materials meet.

(2) Polar moment of Inertia for each comp

$$J_{\text{steel (solid)}} = \frac{\pi d^4}{32} = \frac{\pi (50 \times 10^{-3})^4}{32} = 3.068 \times 10^{-8}$$

$$J_{\text{alu (Hollow)}} = \frac{\pi (d_o^4 - d_i^4)}{32} = \frac{\pi [(76 \times 10^{-3})^4 - (50 \times 10^{-3})^4]}{32} = 9.327 \times 10^{-8} \text{ m}^4$$

(3) Max Torque

As Angle of twist (θ) is same for both

~~Then~~

$$\frac{T_{\text{steel}} L}{J_{\text{steel}} G_{\text{steel}}} = \frac{T_{\text{alu}} L}{J_{\text{alu}} G_{\text{alu}}}$$

$$\frac{T_{\text{steel}}}{T_{\text{alu}}} = \frac{J_{\text{steel}} \times G_{\text{steel}}}{J_{\text{alu}} \times G_{\text{alu}}}$$

$$\frac{T_{\text{steel}}}{T_{\text{alu}}} = \frac{3.068 \times 10^{-8} \times 77 \times 10^9}{9.327 \times 10^{-8} \times 27 \times 10^9} = 0.943$$

Hence $T_{\text{steel}} = 0.993 \cdot T_{\text{alum}}$

(4) Max Torque Determination

By max shear stress

$$\tau_{\text{max}} = T_{\text{max}} \cdot r$$

$$J_{\text{steel}}$$

$$120 \times 10^6 = T_{\text{steel}} \times 0.025$$

$$3.068 \times 10^{-8}$$

$$T_{\text{steel}} = 147.26 \text{ N.m}$$

For Aluminium, $70 \times 10^6 = \frac{T_{\text{alum}} \times 0.038}{9.327 \times 10^{-8}}$

$$T_{\text{alum}} = 171.64$$

Ans

(1) Limiting torque governed by (T_{steel}) = 147.26 Nm

(2) Max torque = 147.26 Nm